METRIC

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MILITARY SPECIFICATION

DETECTING SET, METALLIC-MINE, PORTABLE

This specification is approved for use by the US Army Belvoir Research, Development and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-D-23359F. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification establishes the performance, test and acceptance requirements for the detecting set, metallic-mine, portable.

2. APPLICABLE DOCUMENTS (Not applicable to this specification)

3. REQUIREMENTS

3.1 Description. The detecting set shall consist of a metallic-mine detector packaged in a transport case. The mine detector shall consist of a search head, associated electronic components, a power supply, connecting cables, a collapsible handle and a headset. Throughout this military specification, the mine detector will be referred to as the detector.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subjected to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The detecting set shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the detecting set may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

3.4 System characteristics.

3.4.1 Performance.

3.4.1.1 Sensitivity. The detector shall detect the following targets presented, one at a time, in a plane parallel to the underside of the search head and passing by the point(s) on the search head where a maximum signal is received at the following distances shown in table 1. Point(s) on the search head where a maximum signal is received shall be provided by the contractor.

Table I. Sensitivity targets.

TARGETS	DISTANCES
Aluminum hollow tube, Grade Alloy 5052, 4.45 cm length, 0.64 cm dia., 0.177 mm ±0.05 mm thickness	13.5 cm
PMN-6 striker pin, 308 series (see figure 2)	7.5 cm
M-14 stainless steel pin, 4.75 mm dia. x 4.75 mm long, 308 series	5.0 cm

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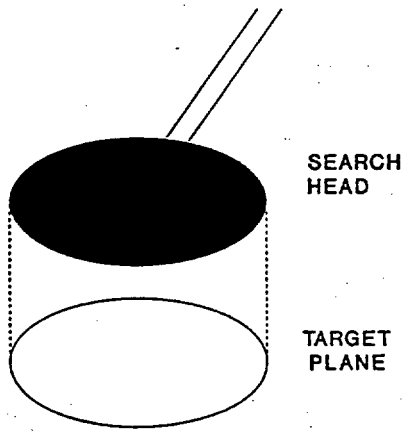


Figure 1. Target zone. X-4949

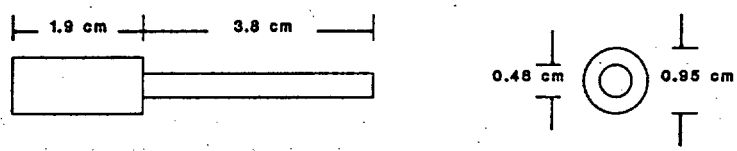


Figure 2. Striker pin. X-4950

3.4.1.2 Detection capability.

3.4.1.2.1 No offset. The detector shall detect the following metallic objects according to the specified depths, soils, and sweep speed. The detector shall detect the targets of table II when the search head receiver coil(s) is at a head height recommended by the contractor. Three of each type of target in table II shall be buried in each soil type for a total of 12 targets per soil type. This shall be referred to as a "lane". A total of 37 passes shall be made over each lane. The properties of the sand, loam and magnetite soils containing moisture from 2 to 25 percent and measured at frequencies of 300 through 600 MHz are as follows:

Sand - Relative dielectric constant of 2.57
Loam - Relative dielectric constant of 3.79
Magnetite - Magnetic susceptibility of 15k x 10^{-6} cgs.

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Table II. Target types and placement.

SOIL SWEEP SPEED TARGETS	Sand 50 cm/sec	Loam 50 cm/sec	Magnetite 50 cm/sec
Aluminum hollow tube, (see 3.4.1.1)	15 cm	15 cm	2.5 cm
PMN-6 striker pin, 308 series (see 3.4.1.1)	10 cm	10 cm	2.5 cm
M-14 stainless steel pin, (see 3.4.1.1)	surface	surface	SURFACE* (SEE NOTE)
M-16 simulated mine	18 cm	18 cm	18 cm

Note: Detection of these targets is desired. Failure to detect these targets will not be counted in the failure criteria (see 4.4.3.4.1). However, if detected, it shall be noted for future reference.

3.4.1.2.2 Offset. The detector shall detect a simulated M-16 mine under the conditions of table II and at a minimum search head offset (from the edge of the search head to the center of the target) of not less than 15 cm (see 4.4.3.4.2).

3.4.1.2.3 Certified mine lanes. The Fort Belvoir RD&E Center certified mine lanes will be made available (see 4.4.3.4a).

3.4.1.3 Environmental operating conditions. The detector shall perform under the following conditions:

a. Soils: Soils of various types shall include clay, dirt, sand and gravel, and magnetite (see 4.4.3.4.1).

b. Water immersion: Water immersion with the detector, including the headset, 1.5 meters below the air-water interface (see 4.4.3.4.4 and 4.4.3.5.2).

c. Sand and dust: The detector shall be resistant to sand and dust (see 4.4.3.5.3).

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d. Temperature:
Operation: -32 ^{0}C to +51 ^{0}C (see 4.4.3.5.1).

e. Transportation: The detector shall be able to withstand the expected modes of transportation including vibration, shock and bounce (see 3.4.2).

f. Humidity: The detector shall be able to withstand the humidity expected in the natural environment (see 4.4.3.5.4).

3.4.1.4 Operational characteristics.

3.4.1.4.1 Required operational characteristics. The detector shall be capable of the following characteristics when employed in a prescribed manner (see 6.4):

a. Having a sweep frequency over a 1.5 meter wide path of 1 complete sweep consisting of across the path, step, and back across the path per 6 seconds (maximum passover time).

b. Operation within 16 meters of another detector of the same model without experiencing or causing electrical interference between detectors (see 4.4.3.10).

c. Detecting but not activating a FFV-028 magnetic influence fuzed mine (see 4.4.3.4.3).

d. Operating without visible or audible emissions which expose the user to enemy detection.

e. Providing not less than 16 hours of continual operation at an ambient temperature of 25 ^{0}C $\pm$5 ^{0}C without degradation of the detector's performance before requiring battery replacement. A battery connection extension kit for body heat battery warming during cold weather operation or other provision may be supplied. Protection against battery installation with reversed polarity is desired. Clear battery polarity markings are required on the batteries and battery casing (see 4.4.3.11).

f. Transmitting the detector output signal to the headset. The signal shall increase in intensity or frequency or both as the detecting head is moved closer to the metallic object.

g. Being operated for at least 20 continuous minutes without a degradation in performance due to aural or physical fatigue. The operator shall be able to resume mine

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detection without degradation in performance after a rest period of no more than 20 minutes (see 4.4.3.12).

h. Being easily operated from either the standing/walking or prone/crawling positions while wearing protective equipment Mission Oriented Protective Posture (MOPP IV), cold weather gear, or helmets and flak jackets (see 4.4.3.12).

i. Detecting mines at the optimum success-rate when the search head is held at a height of 5 to 20 cm and at a sweep speed of 50 cm/sec (see 4.4.3.12).

j. The detector shall be capable of being carried, assembled and operated by personnel wearing Nuclear, Biological, Chemical (NBC) protective equipment Mission Oriented Protective Posture (MOPP IV), cold weather gear, or helmets and flak jackets (see 4.4.3.12).

k. Having a greater than 92 percent probability of detecting standard metalic military mines and mines containing small metallic content buried in various types of soils (see 3.4.1.3a) at depths consistent with doctrinal placement of mines when operated by an MOS-immaterial soldier (see 4.4.3.12).

3.4.1.4.2 Desired operational characteristics. It is suggested that the detector have the ability to locate the center of a target by the x-y coordinate method (see 4.4.3.15).

3.4.1.5 Physical characteristics

3.4.1.5.1 Required physical characteristics. The detecting set shall have the following characteristics:

a. The detecting set shall consist of the following components:

(1) Detector assembly with headset

(2) A transport case/shipping container that shall:

(a) be lightweight and re-usable

(b) be waterproof

(c) have latches that facilitate opening and closing while wearing Nuclear, Biological, Chemical protective equipment (MOPP IV) or cold weather gear.

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(d) contain a pressure relief valve (desired).

b. The detecting set, with its transport case, shall not exceed a total weight of 35 pounds.

c. Controls shall be provided for:

(1) Power ON/OFF

(2) Headset loudness or sensitivity or equivalent automatic adjustment

d. The detector shall be so extendable that the search head can be comfortably held at a sweep-height of 5 to 20 cm above the ground by a standing/walking user whose body is within the 5th percentile female to 95th percentile male. For use in the prone/crawling position, the detector shall be so collapsible that the distance between the base of the search head and the center of the user's hand shall be 0.5 meters or less when naturally gripping the detector.

e. Control for making adjustments of operation for the detector shall be positioned such that the operator can perform any control adjustment in a standing/walking, or prone/crawling position. Controls and displays shall be clearly labeled to indicate the function served by the control or display.

f. The detecting set shall have a non-glare earth tone surface on all external parts (i.e. forest green, brown, black).

g. Each detecting set shall be permanently marked in one location with a serial number assigned by the manufacturer.

h. The detecting set shall not be damaged by any form of moisture (see 4.4.3.5.2).

i. The detector shall be powered by standard dry or alkaline batteries.

3.4.1.5.2 Desired physical characteristics. It is suggested that the detector meet the following characteristics:

a. The field assembly time of the detector shall be 5 minutes or less while wearing NBC protective equipment (MOPP IV), cold weather gear, or helmets and flak jackets.

b. The detecting set shall be operable and maintainable by 5th percentile female through 95th percentile male soldiers.

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3.4.2 Induced environment. The detecting set shall operate as required after exposure to the following induced environmental conditions:

a. Vibration: The detecting set shall be resistant to vibrational stresses expected in shipment and use when tested in accordance with 4.4.3.6.

b. Shock: The detecting set shall withstand infrequent non-repetitive shocks or transit vibrations expected to be encountered in handling and service environments when tested in accordance with 4.4.3.7.

c. Bounce: The detecting set shall not be damaged by bounce when tested in accordance with 4.4.3.8.

3.4.3 Low voltage performance. The detector shall have a low-voltage aural indication. The detector shall exhibit no degradation of detection capability up to the point when the supply voltage decreases to a level that activates the low-voltage aural indication when tested in accordance with 4.4.3.9.

3.4.4 Maintainability. The Mean-Time-To-Repair (MTTR) of the mine detector at the unit level shall be 20 minutes or less with no special tools or equipment required when tested in accordance with 4.4.3.13.

3.4.5 Reliability. The mine detector shall demonstrate a lower test Mean-Time-Between-Failure (MTBF) of 500 hours and an upper test MTBF of 1000 hours when tested in accordance with 4.4.3.14.

3.4.6 Safety.

3.4.6.1 Mechanical safety. Rotating and reciprocating components or components capable of exhibiting temperatures equaling to or greater than 60 °C exposed to contact by personnel shall be guarded/insulated. Detector shall not contain toxic or radioactive materials.

3.4.6.2 Electrical Safety. The detector shall provide for protection of personnel from inadvertently contacting voltages capable of producing shock hazards, insuring that all external parts, surfaces and shields, exclusive of antenna and transmission line are at ground potential at all times during normal operation. Plugs for use with metal cased detectors shall have provision for automatically grounding the metal frame or case of detector when the plug is mated with receptacle, and the grounding pin shall make first contact and break last.

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3.4.6.3 Nonionizing Radiation Safety. Detectors capable of emitting radio frequency radiation between 300 KHz to 100 GHz and having power density greater than or equal to 10 mW/cm2 shall not allow overexposure of personnel via design, fabrication, shielding or method of operating.

3.5 Design and construction.

3.5.1 Workmanship. Workmanship shall be in accordance with routine commercial standards applicable to soldering, wiring, welding and brazing, plating, riveting, finishing, and screw assemblies. All parts, components and assemblies shall be clean and free from dirt, sand, pits, scale, flux, contaminants such as lubricating oils, mold release agents, corrosive products, loose, spattered, or excess solder, weld metal, metal chips and any other foreign material which might detract from the intended operation or function of the equipment. Mounted parts shall be assembled and secured in the specified manner to satisfactorily accomplish the purpose for which they were intended. Parts, such as hinges, catches, handles, and knobs shall be installed so as to avoid damaging the part or mounted surface. Threaded parts such as screws, nuts, bolts, etc., shall show no evidence of cross-threading, mutilation, or detrimental or hazardous burrs.

3.5.2 Interchangeability. All parts bearing the same part number shall be physically, electrically and functionally interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, subject to the approval of the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling

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inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article detecting sets shall be examined as specified in 4.4.2. Each detecting set shall be subjected to the examinations marked "X" in column 2 of table IV. Presence of one or more defects shall be cause for rejection of the first article detecting sets.

4.3.2 First article tests. The first article detecting sets shall be subjected to the tests marked "X" in column 1 of table V. Order of tests is in column 4 of table V. Failure of any test shall be cause for rejection of the first article detecting sets.

4.4 Quality conformance inspection.

4.4.1 Sampling. The sample size for examination shall be determined from table III, column 2. The sample size for test shall be determined by table III, column 3. A sample shall consist of one or more mine detectors drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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Table III. Sample size.

Lot or batch size	Examination sample size	Test sample size
2 to 8	2	2
9 to 15	3	2
16 to 25	5	3
26 to 50	8	5
51 to 90	13	5
91 to 150	20	8
151 to 280	32	13
281 to 500	50	13
501 to 1200	80	20
1201 to 3200	125	32

4.4.2 Examination. Sample detecting sets selected in accordance with 4.4.1 shall be examined for the defects listed in table IV, column 3. Presence of one or more defects shall be cause for rejection of the lot.

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Table IV. Examination schedule.

1	2	3	4
Defects	First article tests	Quality conformance	Requirements
101. Physical characteristics not as specified.	X	X	3.4.1.5
102. Identification markings not as specified.	X	X	3.4.1.5.1g
103. Workmanship not as specified.	X	X	3.5.1
104. Interchangeability not as specified.	X	X	3.5.2
105. Transport case not as specified.	X		3.4.1.5.1a(2)
106. Headset not fitting under standard issue helmet.	X		3.4.1.4.1j
107. Use with NBC gear not possible.	X		3.4.1.4.1j
108. 5 minute assembly with NBC gear not possible.	X		3.4.1.5.2a
109. Batteries not as specified.	X	X	3.4.1.5.1i
110. Headset not immersible.	X		3.4.1.5.1h
111. Material not as specified.	X		3.3
112. Material deterioration prevention and control not as specified.	X		3.3.1
113. Dissimilar metals not as specified.	X		3.3.2
114. Safety	X		3.4.6
115. Soldier compatability	X		3.4.1.5.2b

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4.4.3 Tests.

4.4.3.1 Individual. Each mine detecting set shall be subjected to the test marked "X" in column 2 of table V. Failure of this test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table V. No adjustment shall be made on the mine detector after completion of the individual tests. Failure of any test shall be cause for rejection.

Table V. Test Schedule.

First article test	Quality Conformance		Test	Test paragraph	Requirement paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X		Sensitivity	4.4.3.3	3.4.1.1
X		X	Detection	4.4.3.4	3.4.1.2
X			Temperature	4.4.3.5.1	3.4.1.3d
X			Humidity	4.4.3.5.4	3.4.1.3f
X		X	Vibration	4.4.3.6	3.4.2a
X		X	Shock	4.4.3.7	3.4.2b
X			Bounce	4.4.3.8	3.4.2c
X			Sand & Dust	4.4.3.5.3	3.4.1.3c
X		X	Leak Test	4.4.3.5.2	3.4.1.3b
X		X	Low Voltage	4.4.3.9	3.4.3
X			Proximity interference	4.4.3.10	3.4.1.4b
X			Continual operation	4.4.3.11	3.4.1.4e
X			Operational test	4.4.3.12	3.4.1.4.1g thru 3.4.1.4.1k
X			Maintain-ability	4.4.3.13	3.4.4
X			Reliability	4.4.3.14	3.4.5
X			Center determination	4.4.3.15	3.4.1.4.2a

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4.4.3.3 Sensitivity. Each detector shall be tested to insure operability. The test shall be conducted with a fresh set of batteries or suitable power supply. The headset output (i.e. amplitude, frequency, etc.) shall be measured and recorded by electronic instruments. The contractor shall describe in detail that information which constitutes a positive detection as heard in the headset (i.e. in amplitude, frequency, or other signal) while also specifying the point(s) on the search head where a maximum signal is received. Information shall also be provided on the frequency, amplitude, or other signal when the detector is in the saturated condition.

a. Each detector shall be placed in a test jig fixture, at least 1 meter from metallic objects (excluding those which are part of the detector itself) and other extraneous magnetic/electrostatic fields.

b. The detector shall be powered up using the on-board batteries or suitable power supply and adjusted to the standard operational null condition.

c. Test in accordance with 3.4.1.1. Less than 100 percent detection constitutes failure of this test.

4.4.3.4 Detection capabilities tests. Balance the detector with the search head not less than 1 meter from metallic objects. The detector shall be operational when subjected to the sand, loam, and magnetite test lanes in a controlled environment. Optimal head height shall be provided by the contractor.

a. The Fort Belvoir RD&E Center certified mine lanes will be made available to all potential suppliers on a first come, first served basis. Reservations will be requested by letter or FAX at (703) 781-0065, Attn: STRBE-NDE and are to be received not less than two weeks in advance of required use. Conformation will be by phone or FAX. Information required includes the number of people, how long the mine lanes will be needed, and, if available, security clearances. If there are no security clearances, special escorts will be provided. The mine lanes will contain all the specified targets (table II), at the specified depths, that will be used during the First Article Tests. Extreme care will be used during use to insure that the positioned targets are not disturbed.

Note: If the supplier is from a foreign country, they shall have prior clearance through their respective Embassy to Headquarters, Department of the Army, before access can be granted.

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4.4.3.4.1 Detection tests. The detector shall detect the targets in table II when buried up to, and including, the maximum depths shown. The detector shall detect the targets of table II when the search head receiver coil(s) is at a head height recommended by the contractor. Test in accordance with 3.4.1.2.1. The test shall be considered failed if, in any lane, any target type is not detected a total of seven or more times.

4.4.3.4.2 Offset capability. The detector shall detect a simulated M-16 mine. Test in accordance with 3.4.1.2.2. Failure to detect the M-16 simulated mine, under the conditions of 3.4.1.2.2, shall constitute failure of this test.

4.4.3.4.3 Magnetic influence fuzed mine. A FFV-028 magnetic influence fuzed mine shall be buried flush with the soil-air interface. A flash bulb shall be connected to the mine to indicate actuation of the fuze. The detector shall detect but not activate the mine at a sweep speed of 50 cm/sec and at a head height ranging from 5 cm to 20 cm. Failure to detect before activating the mine shall constitute failure of this test.

4.4.3.4.4 Immersed Operation. The detector shall detect a 4.75 mm dia. x 4.75 mm long mild steel pin when the search head receiver coil is submerged 1.5 m below the water/air interface and is 7.6 cm above the test pin. The test shall be performed in a 3.5 percent synthetic sea-salt solution with a salinity pH value of 8 $\pm$0.2 and at a temperature of 18 ^{0}C $\pm$10 ^{0}C. Any leakage or damage to the detector or failure to detect the mild steel pin shall constitute failure of this test.

4.4.3.5 Environmental.

4.4.3.5.1 High/low temperature operation and thermal shock. Place the disassembled detector in the temperature chamber and reduce the temperature to -32 ^{0}C $\pm$2 ^{0}C. Maintain this temperature for 3 hours, assemble, and remove the detector from the chamber. Test in accordance with 4.4.3.3 using only the PMN-6 striker pin, 308 series (see figure 2), starting within 5 minutes and completing within 30 minutes after removal from the chamber. This test shall be duplicated for six continuous test cycles. After the sixth cycle, test in accordance with 4.4.3.3 using all targets. Failure of 4.4.3.3 or inability to assemble the detector shall constitute failure of the low temperature operation test. Place the disassembled detector in the chamber and increase the temperature to +51 ^{0}C $\pm$2 ^{0}C. Maintain this temperature for 3 hours, assemble, and remove the detecting set from the chamber. Test in accordance with 4.4.3.3, starting within 5 minutes and completing within 30 minutes after removal from the chamber. This test shall be duplicated for six

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continuous cycles. After the sixth cycle, test in accordance with 4.4.3.3 using all targets. Failure of 4.4.3.3 or inability to assemble the detector shall constitute failure of the high temperature operation test.

4.4.3.5.2 Leak test. The detector, with headset, shall be completely immersed in water so that all parts are 1.5 meters below the air-water interface for a period of two hours. The headset shall be removed after 5 minutes of immersion and allowed to dry for the remainder of the two hour immersion test. The temperature of the water shall be 18 ^{0}C $\pm$10 ^{0}C. Bubbles emanating from within the detector shall be considered a leakage. However, bubbles which result from trapped air on the various exterior surfaces of the detector shall not be considered a leak. If leakage is observed, the test shall be discontinued. Any leakage damage to the detector or failure to operate in accordance with 4.4.3.3 shall constitute failure of this test.

4.4.3.5.3 Sand & dust. The sand and dust concentrations shall be 0.177 grams/cubic meter (natural conditions) and tested with a velocity of 8 meters/second. The detector head and electronic package faces shall be oriented toward the blowing sand and dust for a minimum of 90 minutes. Heavy surface erosion/abrasion, penetration of seals, function impairment or failure to operate in accordance with 4.4.3.3 shall constitute failure of this test.

4.4.3.5.4 Humidity. The detector, out of its transport case, shall be subjected to the following humidities along with their corresponding temperatures. This constitutes one cycle. This cycle shall be repeated five (5) continuous times and tested in accordance with 4.4.3.3 after each complete cycle. Failure to operate in accordance with 4.4.3.3 after each cycle shall constitute of this test.

Humidity- %	Temperature-^{o}C	Hours
95	26	6
85	30	2
74	35	8
85	30	2
95	26	6

4.4.3.6 Vibration. The detector, out of its transport case, shall be mounted to the vibration table. It shall be vibrated along each of three orthogonal axes for a period of 1 hour, through a frequency range of 10 to 55 Hz, with a 0.25 cm double amplitude or maximum acceleration of 1.0g with a sweep time of 15.05 minutes logarithmically. The sweep time includes an ascending plus a descending sweep of the frequency range. Any

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physical damage or failure of the detector to operate in accordance with 4.4.3.3 shall constitute failure of this test.

4.4.3.7 Shock. The detecting set shall be loaded, complete with accessories, in the transport case. A total of 26 drops, from a height of 122 cm, on all edges, corners, and faces shall be made onto plywood with a concrete base from a quick release hook or drop tester. The case shall be so oriented that, upon impact, a line from the struck corner or edge to the center of gravity of the case and its contents is perpendicular to the impact surface. Evidence of damage to any detector components or failure to operate in accordance with 4.4.3.3 shall constitute failure of this test. Damage to the transport case shall be allowed provided that it does not prohibit continuing its mission.

4.4.3.8 Bounce. The detecting set shall be loaded, complete with accessories, in the transport case. Test the detecting set on a package-testing machine, with a plywood base, capable of maintaining a constant speed of 260 rpm (4-1/3 cycles per second) with an amplitude of 1.27 cm for a 2.54 cm excursion. Place the detecting set inside the "fence" with 2.54 cm spare or "throw". Bounce the detecting set in the three mutually perpendicular directions; that is, the ends, the sides, and the top and bottom of the detecting set case shall be subjected to the bumping, each for a period of 15 minutes (total time 45 minutes). Any physical damage or failure of the detector to operate in accordance with 4.4.3.3 shall constitute failure of this test.

4.4.3.9 Low voltage performance. Place the mine detector in an ambient temperature of +25 ^{0}C $\pm$5 ^{0}C. Adjust the power source voltage to the detector's standard operating voltage dc, balance the detector in air and reduce the power supply voltage until the low voltage aural indication is actuated. Record the voltage. Then raise the source voltage just enough to stop the low voltage alarm. Test in accordance with 4.4.3.3. Failure to operate in accordance with 4.4.3.3 shall constitute failure of this test.

4.4.3.10 Proximity interference. Place two detectors 16 meters $\pm$10 cm apart such that neither detector is within 1 meter of any metallic object. Turn on and balance each detector. Aural indications in the headset shall be no different than if only one detector was in operation. Repeat above test with one detector in the target detecting mode. Any aural indication in either detectors' headset other than normal presentations, as stated by the contractor, shall constitute failure of this test.

4.4.3.11 Continual operation. Two detectors shall be provided with a fresh set of batteries and balanced, if required, in

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ambient conditions. The detectors shall be placed in saturation with a target placed at the point(s) on the search head where a maximum signal is received. Leave the detectors as such for 16 hours. At the 15 hour 45 minute mark, null the detectors, without changing batteries, and test in accordance with 4.4.3.3. Failure to operate in accordance with 4.4.3.3 shall constitute failure of this test.

4.4.3.12 Operational test. User testing shall assess the ability of MOS-immaterial soldiers to operate, support and maintain each of the commercial AN/PSS-12 mine detector candidates in an operational environment during daytime and night time. Operator and maintenance training shall be conducted in accordance with the training plan for each type of mine detector. Mine detectors shall be employed in a prescribed manner (see 6.4). The detector shall have a greater than 92 percent probability of detecting standard metallic military mines and mines containing small metallic content buried in various types of soils at depths consistent with doctrinal placement of mines when operated by an MOS-immaterial soldier (see 3.4.1.4.1k). The distance from the ground to the bottom of the head when the detector is employed shall be approximately 5 to 20 cm. Mine lanes shall require the mine detector to be used in areas of long grass, underbrush, leaves and stones, and with the detector head submerged in water. The test shall evaluate the ability of soldiers to use the mine detectors when walking, standing, or in the prone/crawling position while wearing cold weather gear as well as mission-oriented protective posture (MOPP) IV gear. Testing shall be conducted in existing weather conditions. During testing, maintenance and system support package adequacy data shall be collected and assessed. Machine/operator interface shall be of special interest during testing. Of particular concern is: the period of time an operator can continuously operate the mine detector before having to be replaced; the ease by which the operator is able to locate mines; the number of times the operator must adjust the null; and the ease of transporting, configuring, operating, and maintaining the detector.

4.4.3.13 Maintainability. Maintenance operations required during the operational testing shall be used to determine conformance to 3.4.4. All maintenance actions performed during any testing (including scheduled maintenance) shall be timed and recorded in order to determine conformance to 3.4.4. For each maintenance action, a complete record of the maintenance action required, the date and time of the occurrence, the cause for the maintenance action, and the time to repair the detector shall be recorded.

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4.4.3.14 Reliability. The lower test Mean-Time-Between-Failure (MTBF) shall be 500 hours; the upper test MTBF shall be 1000 hours. Three detectors shall accumulate a total of 1920 hours of operation as described below. Each detector tested shall operate at least one-half the average of all mine detectors tested. An external power supply may be used. The following cycle shall be repeated until the total number of operating hours are completed: The detectors shall be turned off and on once every twenty minutes until 9 hours of operation are accumulated. The power off period shall be at least 30 seconds and shall not exceed one minute. The detector shall then be subjected to fifteen (15) hours of continuous operation followed by a sensitivity test in accordance with 4.4.3.3. Additionally, the search head of one mine detector shall be moved through its entire range of extension and resecured a total of 720 times (the same detector/head is to be exercised for the duration of the test). All failures will be scored by Government personnel. For purposes of determining the MTBF, a failure is defined as any malfunction, degradation, or performance outside specified limits which the operator cannot remedy by adjustment within 5 minutes using tools or parts supplied with the detector; or a hazard to personnel or equipment. More than two failures when tested in accordance with 4.4.3.3, loss of the ability to adjust or secure the search head handle or failure of the on/off switch shall constitute failure of this test.

4.4.3.15 Center determination. The detector shall determine the center of a target by using the x-y coordinate method. For targets less than or equal to 7.5 cm in diameter, the detector shall determine the center of the target within ±7.5 cm. For targets greater than 7.5 cm, the detector shall determine the center of the target within ±15 cm. Figure 3 illustrates this method. This is a desired feature and, therefore, has no failure criteria. However, the ability to accomplish this test will be noted for future reference.

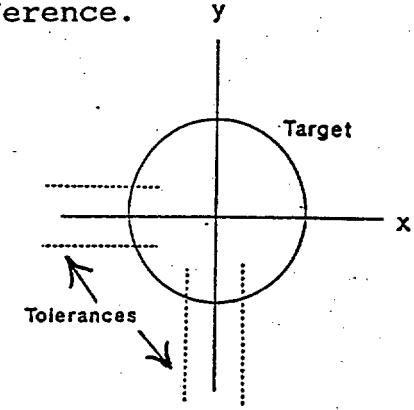


Figure 3. Center determination.

X-4951

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4.5 Inspection comparison. The Government may select detecting sets at any time during the contract production period and subject these detecting sets to the examination and tests specified, to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5.1 Inspection failure. Failure of an inspection comparison detecting set to meet any requirement specified herein and as a result of the examination and tests specified in 4.5 shall be cause for rejection of the inspection comparison detecting set(s) nd shall be cause for refusal by the Government to continue acceptance of production detection sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on detectors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all detecting sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. The sample size shall be determined from table III. A sample shall consist of one or more units drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects.

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201. Preservation not as specified (see 5.1).
202. All components of a complete detecting set not placed in the transport case as specified (see 5.1.2).
203. Transport case not closed or secured as specified (see 5.1.2).
204. Packing not as specified (see 5.2).
205. Marking not as specified (see 5.3).

5. PACKAGING.

5.1 Preservation. Each detecting set and all of the components shall be preserved in an appropriate manner to afford protection against deterioration and damage from the contractor to the initial destination. The contractor may use their commercial practice provided it fulfills these requirements.

5.1.2 Consolidation. All components of a complete detecting set shall be placed in the transport case. The case shall be closed and secured.

5.2 Packing. Each detecting set, preserved as specified in 5.1, and consolidated in the transport case, shall be packed in wooden boxes that will provide suitable protection against damage during shipment and handling from the contractor to the initial destination. Quantities of sets packed per box shall not exceed 100. Appropriate blocking and bracing shall be utilized to prevent movement and damage.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with military standards.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. This specification covers a detecting set intended for use over all types of terrain for the location of buried metallic antipersonnel and antitank mines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Whether batteries are to be furnished.

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c. Time frame required for submission of the first article models and number of detecting sets to be furnished. Standard samples required (see 6.3).
d. Amount of preservation and packing required (see 5.1 and 5.2).
e. Quantity of packed detecting sets required to be consolidated (see 5.2).
f. The number of detecting sets composing a lot.
g. Any special marking (see 5.3).

6.3 First Article. When a first article inspection is required, the item(s) should be production models. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Standard sample. Standard sample detecting set shall include a transport case, initial set of batteries, technical/operators manual, maintenance/repair manual, and all other components as specified in 3.1. The standard sample shall be representative of a production lot of the specific detector model; i.e., it shall be identical in design, construction, and performance. The quantity of detecting sets to constitute the standard sample will be specified in the contract or order.

6.4 Operating Manual. The mine detector shall be employed as prescribed in FM 20-32. This manual will be furnished at the contractor's request.

6.5 Subject term (key word) listing.
Detection
Sensitivity

Custodian:
Army-ME

Preparing activity:
Army-ME

Project 6665-A606

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-D-00233596 (ME)	90/03/23

3. DOCUMENT TITLE
Detecting Set, Metallic - mine, Portable

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466	

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